Ahmed Iprahim Abdelhamed

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with production experience in quantitative analysis, feature engineering, and statistical modeling within fintech (Tabby). Proven expertise in architecting end-to-end machine learning pipelines across supervised classification, deep tabular neural architectures, and collaborative filtering. Skilled in training, tuning, and deploying models using Python, Scikit-Learn, TensorFlow / Keras, and PyTorch to solve real-world challenges like severe class imbalance and data leakage.

CORE TECHNICAL SKILLS

Machine & Deep Learning: XGBoost, Scikit-Learn, SVD, TensorFlow / Keras (Functional API, Residual Blocks), PyTorch (Tensors, Autograd), Hyperparameter Optimization.

Feature Engineering: ColumnTransformer, Smooth Target Encoding, Cascade Imputation, Missingness Analysis (MAR/MCAR), Multicollinearity Diagnostics (VIF).

Languages & Tooling: Python (NumPy, Pandas, SciPy, Statsmodels), SQL (PostgreSQL, DuckDB), Git, Bash.

Evaluation & Deployment: Precision-Recall AUC, ROC-AUC, RMSE, R^2, Streamlit Web Applications, Cross-Validation.

PROFESSIONAL EXPERIENCE

Tabby — Data & Analytics Specialist (Fintech Operations)

2023 – Present | Cairo, Egypt

- **Automated ML Pipelines:** Engineered Python data pipelines to extract, clean, and validate multi-source event logs (50,000+ daily records), establishing feature pipelines and anomaly checks that cut manual processing overhead by 80%.
- **Risk Escalation & Fraud Investigations:** Queried relational databases to identify recurring delinquency patterns, escalating behavioral anomalies to Credit Risk teams. Investigated unauthorized disputes and Account Takeovers via internal admin tools.

Independent Practice — Data Scientist & Risk Analyst (Freelance)

2023 – Present | Remote

- **Applied Machine Learning:** Delivered comprehensive end-to-end data pipelines and predictive analytics solutions for financial lending datasets.
- **ETL & Data Architecture:** Architected Kimball Star Schema data models and automated ETL workflows to ingest, clean, and consolidate disparate transaction datasets for downstream machine learning applications.

MACHINE LEARNING & DEEP LEARNING PROJECTS

Large-Scale Vehicle Valuation Engine (Deep Tabular ResNet)

Tech Stack: TensorFlow / Keras, Python, Scikit-Learn (ColumnTransformer, TargetEncoder), Missingno

- Architected a custom Deep Tabular ResNet using Keras Functional API with residual skip connections and bottleneck projections, achieving an R^2 of 0.777 on 360,000+ real-world vehicle listings.
- Designed a 3-tier cascade imputation engine leveraging Missing at Random (MAR) analysis and domain heuristics (e.g., zero-cylinder mapping for EVs) to eliminate nested missingness.
- Built a multi-stage preprocessing pipeline integrating smoothed Target Encoding, One-Hot Encoding, and RobustScaler to eliminate target skewness and prevent train/val/test data leakage.

Credit Risk & Expected Loss AI Engine

Tech Stack: Python, XGBoost, Scikit-Learn, PostgreSQL, Streamlit

- Engineered a credit classification pipeline evaluating Probability of Default (PD) and Expected Loss across 32,000+ borrower records, strictly preventing data leakage by excluding post-outcome features.
- Tackled severe class imbalance (78:22) by optimizing XGBoost scale_pos_weight and tuning hyperparameters, increasing default recall to 78% without compromising precision.
- Deployed the predictive model as an interactive Streamlit web app, enabling instant applicant scoring, automated risk tier assignment, and real-time simulations for underwriters.

CineMatch AI – Hybrid Recommender System

Tech Stack: Python, Scikit-Learn, NumPy, Pandas, Streamlit

- Developed a hybrid recommendation engine processing 10,000,000+ MovieLens ratings using Singular Value Decomposition (SVD) with user/item bias and latent factor modeling.
- Attained an RMSE of 0.8794 (outperforming baselines by 17%) and deployed a responsive Streamlit app featuring fuzzy string matching and real-time cosine similarity search.

EDUCATION & CERTIFICATIONS

Bachelor of Science in Computer Science — Helwan University | Graduated 2024

Certifications: Advanced AI & Deep Learning Specialization (In Progress, 2026) | ALX Data Science, Machine Learning, Data Analytics & Python Programming Certificates (2024).